Supplementary Material for "Who Bears the Burden of Detecting AI with AI? A Model of Collateral Performative Prediction in the Detection Arms Race"

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This supplement reports the robustness checks and protocol details that the main paper summarises. Section, proposition and table numbers without an "S" prefix refer to the main paper. Unless stated otherwise, the conventions of Section 5.1 of the main paper apply: values are measured after 60 rounds, averaged over independent seeds, and intervals describe variability across seeds under fixed parameters.

S1 Additional Simulation Results

S1.1 Sensitivity analysis

Table S1 varies one parameter at a time around the defaults, with 10 seeds per setting and the default row from 20 seeds. Its initial disparity rests on few flagged texts and is noisy: the default is 9.7 over 20 seeds, 11.8 over the first 10, and 12.7 when computed without sampling. In every setting, AUROC falls below 0.75 within the 60 rounds, ending between 0.50 and 0.72; the minority group drifts further and accumulates more expected false flags per writer; and initial disparity grows with the group offset m_2 (3.5 to 50.5 times) and with stricter operating points (6.5 times at 5%, 22.0 times at 0.1%), as Proposition 4.6 predicts. As avoidance strengthens from $\lambda_h = 0$ (the no-adaptation ablation, in which homogenisation is also switched off) through 0.25, 0.5 and 1.0, final AUROC rises from 0.54 to 0.55, 0.61 and 0.69, while G2's drift rises from 0.07 to 0.93, 1.73 and 3.23: within the default model, the detector's residual accuracy is bought with human style change. A 0.1% operating point slows the loss of AUROC, because fewer machine texts are caught and evasion pressure is weaker, at the price of the largest initial disparity and drift, although each G2 writer then accumulates only 0.11 expected false flags rather than 0.82; its realised rate is 0.15%, because a 2,000-text calibration sample has few observations in that tail.

Table S1: One-at-a-time sensitivity analysis around the default parameters.

Param.	Value	Init. disp.	Final AUROC	Below 0.75	Drift G1	Drift G2	Flags G1	Flags G2
Default		9.7	0.61	17	0.94	1.73	0.58	0.82
λ_h	0.25	11.8	0.55	15	0.24	0.93	0.56	0.90
	1.0	11.8	0.69	27	2.38	3.23	0.60	0.75
η	0.3	11.8	0.67	27	0.83	1.62	0.59	0.83
	1.2	11.8	0.54	9	1.03	1.80	0.58	0.81
ρ	0.05	11.8	0.57	11	1.01	1.78	0.58	0.81
κ	0.03	11.8	0.58	13	0.25	1.11	0.59	0.76
m_2	0.5	3.5	0.61	17	1.02	1.41	0.60	0.72
	1.5	50.5	0.61	17	0.85	2.03	0.56	0.92
a_0	2.0	10.6	0.61	15	0.98	1.75	0.58	0.81
	4.0	11.9	0.61	17	0.90	1.69	0.58	0.82
α	0.1%	22.0	0.72	50	1.39	2.16	0.09	0.11
	5%	6.5	0.50	7	0.68	1.47	2.81	3.83

S1.2 Interaction between avoidance and evasion

Figure S1 sweeps avoidance $\lambda_h \in \{0, 0.25, 0.5, 1\}$ and the evasion step $\eta \in \{0.15, 0.3, 0.6, 1.2\}$ jointly, with 10 seeds per cell. As in the sensitivity analysis, the $\lambda_h = 0$ row is the no-adaptation ablation, with homogenisation also switched off; it therefore matches "No avoidance or homogenisation" (final AUROC 0.54 at $\eta = 0.6$) rather than the avoidance-only ablation of Section S1.4, which keeps homogenisation and ends at about 0.50. With strong avoidance and weak evasion ($\lambda_h = 1$, $\eta \leq 0.3$), final AUROC remains above 0.75 (0.82 and 0.76), but these are the cells in which G2's style changes the most (drift 2.91 and 3.08). Final AUROC increases with λ_h in 11 of 12 adjacent comparisons and decreases with η at every level of λ_h. G2 writers accumulate more expected false flags than G1 writers in all 16 cells, and G2 drifts further whenever $\lambda_h > 0$; without avoidance G1 drifts more under assimilation, and the disparity appears in false flags instead (1.83 to 1.94 per G2 writer against 0.30 to 0.34).

S1.3 Robustness to noisy feedback

The group rates that drive the human update are estimated from finite samples, about 400 texts for G2. Because the score is linear and each population Gaussian, the threshold, true-positive rate and group rates can also be computed exactly for the fitted detector. With exact rates, final AUROC changes by at most 0.004 in the default and anchored settings, the round at which AUROC first falls below 0.75 is unchanged, and group drift changes by at most 0.08 units; a five-fold larger sample gives a final AUROC of 0.610. Only the initial disparity ratio changes (12.7 exact against 9.7 sampled), because the ratio of two small, noisy rates is biased downward.

S1.4 Separating assumed responses from feedback

Table S2 reports the paired differences from the full loop on which Section 5.3 draws, with 95% t-intervals over the 20 simulated seeds. "Neither" removes avoidance and homogenisation with assimilation still active; "No human change" also removes assimilation; the open loop freezes each group's pressure and direction at their round-0 values. Assimilation alone lowers final AUROC by 0.045 compared with no human change.

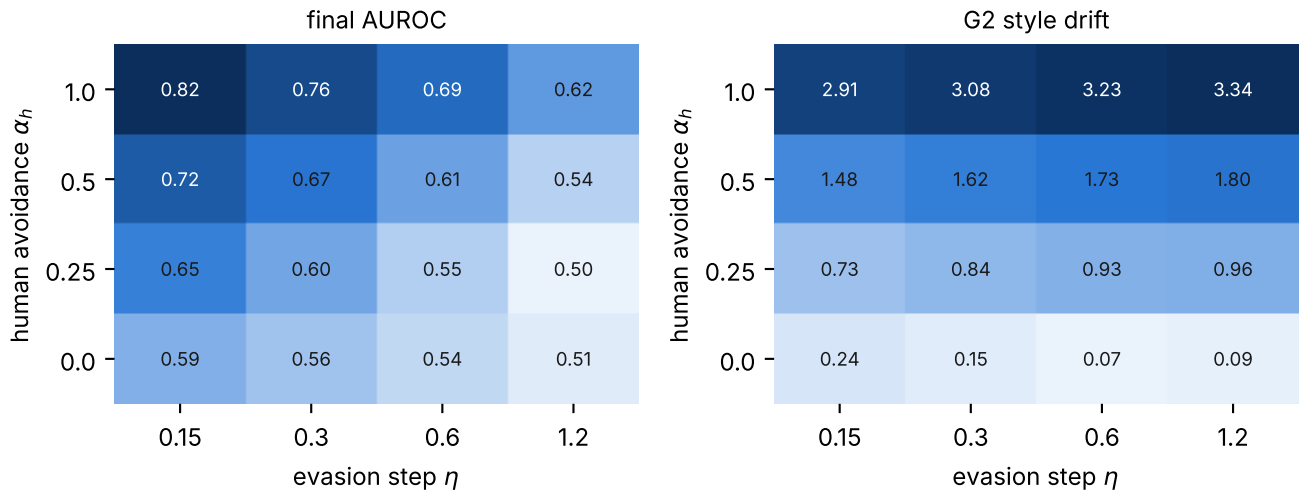


Figure S1: Joint sweep over human avoidance strength λ_h and evasion step η after 60 rounds (mean of 10 seeds per cell). Left: final detector AUROC. Right: style drift of the minority group G2.

Table S2: Paired differences from the full loop: separate ablations and the open-loop control.

Configuration	Δ AUROC	Δ Drift G1	Δ Drift G2	Δ Flags G1	Δ Flags G2
No avoidance	−0.107 [−0.112, −0.102]	−0.55 [−0.58, −0.51]	−1.66 [−1.70, −1.62]	−0.06 [−0.07, −0.05]	+0.23 [+0.22, +0.24]
No homogenisation	+0.073 [+0.069, +0.076]	−0.11 [−0.12, −0.10]	+0.09 [+0.08, +0.10]	−0.02 [−0.03, −0.02]	+0.10 [+0.09, +0.10]
Neither	−0.070 [−0.074, −0.066]	−0.48 [−0.51, −0.44]	−1.67 [−1.71, −1.62]	−0.27 [−0.27, −0.26]	+1.07 [+1.04, +1.10]
No human change	−0.025 [−0.028, −0.022]	−0.95 [−0.98, −0.91]	−1.73 [−1.77, −1.69]	−0.38 [−0.39, −0.37]	+1.53 [+1.49, +1.57]
Open loop	+0.006 [−0.000, +0.011]	−0.68 [−0.76, −0.60]	+2.48 [+2.00, +2.97]	+0.14 [+0.13, +0.15]	−0.55 [−0.59, −0.51]

S1.5 Less idealised human responses

The default rule has writers move exactly away from the deployed detector, at once and without limit. We replace it with five less idealised versions, with the default parameters otherwise (Table S3). *Noisy*: the direction of response is $\hat{w}_t + \sigma\xi_t$, normalised, with ξ_t a fresh standard normal vector scaled by $1/\sqrt{d}$; $\sigma = 1$ and 3 give angular errors of about 45 and 70 degrees. *Misdirected*: writers move away from a different detector, along $\cos\phi\,\hat{w}_t + \sin\phi\,v_t$ with v_t a fixed random direction orthogonal to $\hat{w}_t$ and ϕ the angle of Definition 3.2; at $\phi = 135°$ a rewrite that lowers the other detector's score raises the deployed one's. *Bounded*: each group's total avoidance displacement is capped at B style units. *Delayed*: the response in round t uses the pressure and direction of round $t - k$. *Heterogeneous*: G2's avoidance and homogenisation rates are scaled by 0.5 or 2.

G2 accumulates more expected false flags per writer than G1, with a paired interval excluding zero, in every variant except a doubled G2 response, in which G2 escapes the flagged region faster (0.49 against 0.66). G2 drifts further than G1 in every variant except orthogonal misdirection, where the two drift equally (1.12 against 1.14). Final AUROC is lower than under the default rule for every noisy, misdirected and bounded response, by 0.016 to 0.105, slightly higher with delays, and within 0.003 of it with heterogeneous responses. Bounded responses trade one cost for the other: G2 changes its style less (0.48 units with $B = 0.5$) but accumulates more false flags (1.08).

Table S3: Less idealised human responses.

Response	Final AUROC	Below 0.75	Drift G1	Drift G2	Flags G1	Flags G2
Default rule	0.61	17	0.94	1.73	0.58	0.82
Noisy, $\sigma = 1$	0.58	15	0.58	1.30	0.58	0.87
Noisy, $\sigma = 3$	0.53	14	0.23	0.69	0.56	0.94
Misdirected, 45°	0.59	15	1.26	1.38	0.61	0.72
Misdirected, 90°	0.55	13	1.14	1.12	0.63	0.66
Misdirected, 135°	0.52	11	0.89	1.79	0.51	1.10
Bounded, $B = 0.5$	0.50	15	0.07	0.48	0.52	1.08
Bounded, $B = 1$	0.52	17	0.53	0.87	0.52	1.09
Delayed, $k = 3$	0.61	18	0.86	1.65	0.58	0.82
Delayed, $k = 10$	0.63	20	0.67	1.34	0.58	0.83
G2 response ×0.5	0.61	17	0.66	1.32	0.48	1.25
G2 response ×2	0.61	17	1.16	2.07	0.66	0.49
Anchored parameters, $r = 1$:						
Default rule	0.50	9	0.16	1.50	0.54	0.99
Misdirected 90°	0.51	9	0.85	1.07	0.55	1.00
Misdirected 135°	0.51	10	0.72	1.17	0.41	1.51

S1.6 A nonlinear detector and several generators

We replace the linear detector by a neural network with one hidden layer of 32 tanh units, trained with Adam and early stopping on the same samples. Responses follow the score's *local* gradient: each human group moves along the mean normalised gradient at its own flagged texts, and each generator along the mean normalised gradient at its caught texts; for a linear detector this reproduces the default model. In a second variant, a second generator starts at $a_0 e_2$, so machine text forms two clusters and the optimal boundary is far from a hyperplane (Table S4, 20 seeds per configuration).

The human-side results carry over: G2 drifts further and accumulates more expected false flags per writer (paired difference 0.19 to 0.27 for the nonlinear detector, every 95% interval excluding zero), and flags pass from G2 to G1, whose false-positive rates move from 3.4% and 0.4% to 1.14% and 1.15% with one generator. Under the nonlinear detector the groups also move in different directions: with two generators, the directions of G1 and G2 make an angle of about 48° at round 0 and still differ by 26° on average over the run. The detector-side result does not carry over: adaptation raises final AUROC by 0.070 and 0.081 with the linear detector, but changes it by −0.010 (−0.031 to +0.012) and +0.005 (−0.025 to +0.036) with the nonlinear one.

S1.7 Anchors from each commercial detector

For each detector of Section 6, we derive m_2 from its native and non-native false-positive rates, δ_h from the polished non-native essays, and δ_A from the GPT-3 texts before and after rewriting, with a continuity correction $(k+0.5)/(n+1)$ because some detectors flag no native essays, and $a_0 = 4.0$ (Table S5). The paired 95% interval for the effect of full response on final AUROC includes zero only for Crossplag. AUROC falls below 0.75 within 10 to 18 rounds with $r = 1$ and within 11 to 30 with $r = 0$; with $r = 0$, G2 accumulates 2.06 to 3.09 expected false flags against 0.01 to 0.28. The Krishna-based anchors give the same picture with δ_A in machine-score units ($\eta = 1.77$; final AUROC 0.50) and with rewriting efficacy that halves every 10 or 30 rounds (0.53 and 0.50).

S1.8 Robustness of the policy comparison

Table S6 tests the policy comparison. For the fixed threshold, we retrain on a machine share π_{tr} other than one half: constant at 0.8, or drifting from 0.5 to 0.8 or to 0.98 over the 60 rounds. For withdrawal, we vary the assumed prevalence π, let it drift, replace the precision rule with a rule that withdraws once audited AUROC falls below 0.75, and audit on 200 labelled texts per class; all withdrawal rules here use point estimates, and Table 10 of the main paper reports confidence-bound audits. The *Platt-recalibrated threshold* recalibrates each retrained model's log-odds by Platt scaling on a fresh balanced labelled sample and holds the cut-off fixed on that probability. As a reference we also apply a cut-off fixed at round 0 (about 0.94) to the exact equal-prior posterior of the populations of the recalibrated runs, computed from their Gaussian densities, with TV_t estimated by Monte Carlo. Its false-positive rate never exceeds the bound of Proposition 4.4(2) in any of the 1,200 seed-rounds per setting, and neither does that of the Platt-recalibrated threshold. The deployed probabilities are poorly calibrated against the equal-prior posterior once the populations move. Their mean absolute difference from the exact posterior, averaged over human and machine evaluation texts, rises from 0.05 at round 0 to 0.18 at round 60 with the default parameters, and from 0.03 to 0.20 with the anchored ones; with a training prior drifting to 0.98 it reaches 0.46. Platt recalibration removes the effect of the training prior (0.19 at round 60 under the same drift) but not the rest of the error, which lies in the shape of the linear score rather than its scale: the log-likelihood ratio of a two-group human mixture against a narrower machine distribution is not linear. Both fixed cut-offs nevertheless keep the false-positive rate within the bound of Proposition 4.4(2) in every seed-round when the training prior is stable or the scores are recalibrated. The raw cut-off exceeds the bound only when the training prior drifts without recalibration, in 25% of seed-rounds.

Every withdrawal variant ends deployment, after 4.6 to 26.0 rounds on average with the default parameters and 1.5 to 9.4 with the anchored ones, and the saving shrinks as the rule becomes more tolerant: assuming a higher prevalence, or using the AUROC rule, keeps a weaker detector in use longer. When the prevalence is unknown or drifting, assuming the lowest plausible value withdraws earliest and is the conservative choice for writers. With a training prior drifting to 0.98, the raw fixed threshold's realised false-positive rate falls below 0.3% and then rises, passing 1% around round 40 and reaching 21% by round 56 (33% with the anchored parameters).

S1.9 Withdrawal under a misspecified or estimated prevalence

This analysis assumes an exact audit, the assumption of Corollary 4.5; Section S1.11 reports withdrawal with a sampled, independent audit. Before withdrawal, a detector under the withdrawal policy follows the same trajectory as under recalibration, and afterwards it flags nothing. For each seed we can therefore compute, from the recalibration trajectory, when the rule withdraws under any assumed prevalence $\hat{\pi}$, how many rounds it stays in use while the true precision is below 1/2 ("bad rounds"), and how many expected false flags per G2 writer it issues relative to an oracle rule that uses the true prevalence π (Table S7, default parameters). The last column gives the lowest true precision observed in use, with the bound of Corollary 4.5 in brackets; it is respected in every cell. Underestimating the prevalence withdraws earlier, at the price of detection lost while flags would still have been informative (5.9 rounds on average when $\hat{\pi} = 0.05$ and $\pi = 0.1$). With the anchored parameters the oracle withdraws after 1.6 rounds, and $\hat{\pi} = 0.3$ keeps the detector for 7.7, adding 0.31 false flags per G2 writer.

When the prevalence is estimated by adjusted classify-and-count, $\hat{\pi} = (\text{flag rate} - \widehat{\text{FPR}})/(\widehat{\text{TPR}} - \widehat{\text{FPR}})$ [1, 2], the denominator is the detector's utility, so the estimate becomes unstable exactly when the decision matters. For an operator who sees 2,000 submissions per round at a true prevalence of 0.05, 0.1 or 0.2 and audits 200 labelled texts per class, the point estimate gives at most 0.35 bad rounds on average with the default parameters, but about one with the anchored ones at $\pi = 0.05$; the lower 5% bound from a parametric bootstrap reduces this to at most 0.15 in every setting, at the cost of earlier withdrawal.

Table S4: Linear and nonlinear detectors, with one or two generators, with and without human adaptation (20 seeds).

Detector	Generators	Initial AUROC	Initial disparity	Final AUROC	Below 0.75	Drift G1	Drift G2	False flags G1	False flags G2
Linear	1	0.981	9.7	0.607	16.4	0.94	1.73	0.58	0.82
Nonlinear	1	0.982	9.5	0.612	14.7	0.90	1.70	0.57	0.84
Linear	2	0.933	5.8	0.608	14.7	1.03	1.57	0.60	0.75
Nonlinear	2	0.970	4.4	0.612	9.6	0.99	1.54	0.59	0.78
No adaptation:									
Linear	1	0.981	9.7	0.536	14.6	0.47	0.07	0.32	1.89
Nonlinear	1	0.982	9.5	0.621	13.3	0.40	0.06	0.42	1.47
Linear	2	0.933	5.8	0.527	11.1	0.37	0.29	0.43	1.40
Nonlinear	2	0.970	4.4	0.607	8.9	0.27	0.27	0.52	1.05

Table S5: Anchors derived from each commercial detector separately, and the simulation they produce.

Detector	m_2	δ_h	δ_A	Final AUROC $r=0$	Final AUROC $r=1$	Drift ($r=1$) G1	Drift ($r=1$) G2	False flags ($r=1$) G1	False flags ($r=1$) G2
GPTZero	2.58	0.92	1.31	0.59	0.50	0.26	1.45	0.44	1.37
OriginalityAI	2.82	1.84	0.88	0.61	0.55	0.19	2.33	0.46	1.28
HFOpenAI	1.51	1.86	1.25	0.53	0.52	0.54	1.66	0.55	0.96
Crossplag	1.17	1.81	1.19	0.52	0.52	0.58	1.44	0.57	0.89
Quil	1.97	1.71	1.02	0.56	0.53	0.35	1.80	0.52	1.09
Sapling	2.11	1.56	1.52	0.56	0.50	0.23	1.73	0.52	1.09
ZeroGPT	2.49	0.92	1.69	0.57	0.50	0.23	1.42	0.46	1.30

Table S6: Robustness of the policy comparison to prior shift, withdrawal rule, audit size and prevalence (20 seeds). Withdrawal rows use point estimates of precision on an audit sample drawn independently of the evaluation sample (Section S1.11).

Parameters	Policy	Rounds deployed	Mean FPR	False flags G1	False flags G2
Default	Fixed FPR (reference)	60	1.05%	0.58	0.82
	Fixed threshold, $\pi_{tr} = 0.5$	60	0.17%	0.06	0.25
	Fixed threshold, $\pi_{tr} = 0.8$	60	0.17%	0.07	0.26
	Fixed threshold, π_{tr}: $0.5 \to 0.8$	60	0.23%	0.09	0.30
	Fixed threshold, π_{tr}: $0.5 \to 0.98$	60	3.90%	2.28	2.60
	Platt-recalibrated threshold, π_{tr}: $0.5 \to 0.98$	60	0.16%	0.06	0.25
	Withdrawal, $\pi = 0.1$	8.4	0.15%	0.05	0.24
	Withdrawal, $\pi = 0.05$	4.6	0.08%	0.02	0.16
	Withdrawal, $\pi = 0.25$	26.0	0.46%	0.23	0.47
	Withdrawal, π: $0.05 \to 0.3$	6.2	0.11%	0.03	0.20
	Withdrawal, AUROC < 0.75	16.3	0.29%	0.13	0.36
	Withdrawal, audit of 200 per class	6.6	0.12%	0.04	0.21
Anchored	Fixed FPR (reference)	60	1.05%	0.52	1.08
	Fixed threshold, π_{tr}: $0.5 \to 0.8$	60	0.14%	0.01	0.40
	Fixed threshold, π_{tr}: $0.5 \to 0.98$	60	5.16%	3.00	3.46
	Platt-recalibrated threshold, π_{tr}: $0.5 \to 0.98$	60	0.12%	0.00	0.34
	Withdrawal, $\pi = 0.1$	1.5	0.03%	0.00	0.08
	Withdrawal, audit of 200 per class	1.6	0.03%	0.00	0.07
	Withdrawal, AUROC < 0.75	9.4	0.17%	0.01	0.46

S1.10 Deployment requirements of the operating policies

Table S8 summarises, for each operating policy compared in the main paper, what it requires in deployment and the failure modes that the analysis and the simulation identify. None of the policies is free of assumptions; withdrawal is the most robust in the model, but it depends on labelled audit data that represent current submissions and current evasion, and at a 1% operating point a confidence-bound

Table S7: Withdrawal with a wrong prevalence.

π	$\hat{\pi}$	Rounds in use	Bad rounds	Extra flags G2	Lowest PPV
0.05	0.05	4.2	0.0	0.00	0.57 (0.50)
	0.1	8.6	3.9	+0.10	0.36 (0.32)
	0.3	38.5	33.8	+0.45	0.12 (0.11)
0.1	0.05	4.2	0.0	−0.10	0.74 (0.68)
	0.1	8.6	0.0	0.00	0.54 (0.50)
	0.2	17.7	7.8	+0.13	0.34 (0.31)
	0.3	38.5	28.6	+0.35	0.23 (0.21)
0.2	0.1	8.6	0.0	−0.13	0.73 (0.69)
	0.3	38.5	11.9	+0.22	0.40 (0.37)

audit needs about 2,000 human texts to bound the false-positive rate.

S1.11 Withdrawal with an independent audit sample

All withdrawal results use a fresh audit sample each round, drawn from the current populations with its own random stream, so that the trajectory before withdrawal does not depend on the audit. Table S9 compares this with an audit that reuses the round's evaluation sample, in which the decision to withdraw and the reported rates share their random draws. The mean withdrawal round changes by at most 1.7 rounds and G2's cumulative false flags by at most 0.03, every paired 95% interval for the change in withdrawal round includes zero, and so does every interval for the change in G2's false flags except one (assumed prevalence 0.05: +0.017, interval +0.000 to +0.033). Withdrawal timing varies across seeds, most with an assumed prevalence of 0.25 (rounds 14 to 33) and with an audit of 200 texts per class (rounds 3 to 11).

S1.12 A constant evasion step

In the main model the evasion step is $\eta\, q\, \mathrm{TPR}_t$, so any rise in the true-positive rate strengthens evasion. To separate the effect of human adaptation from this rule, we hold the step at $\eta\, q\, c$ with a constant rate c, set to the mean true-positive rate over rounds and seeds of either the matching no-response run or the matching response run, and compare the response with no response under the same c (Table S10). "No response" switches off avoidance and homogenisation; "response" is the default rule, or $r = 1$ with anchored parameters. Under anchored parameters, avoidance no longer changes final AUROC once the step is constant, so the acceleration of Section 5.4 of the main paper operates through the catch rate. With default parameters, the effect of adaptation on accuracy depends on how strong evasion is. In every setting, G2's cumulative false flags fall and G1's rise when writers respond, G2 still accumulates more false flags than G1, and G2's style drifts further.

S1.13 Alternative response functions

The main model maps each group's pressure $g = \mathrm{FPR}_{g,t}/\alpha$ linearly to its avoidance and homogenisation steps. Table S11 replaces this mapping by a threshold, under which a group responds only while g is at least 2 or 5; a saturating function $c\tanh(g/c)$ with cap $c = 2$ or 1; and a linear response scaled to a quarter, as if only a quarter of flagged writers learned of their flag. At round 0 G2's pressure is about 4 and G1's below 1, so under the 5× threshold neither group ever responds, and only assimilation moves human style. G2 accumulates more false flags than G1 in every configuration, with every paired interval excluding zero. Flags pass from G2 to G1, so that G2's final rate falls to or below G1's, under the linear response, the 2× threshold and every anchored variant, but not under saturating or partial responses, which are too weak for G2 to leave the flagged region within 60 rounds.

S1.14 Imperfect recalibration

Under the fixed-FPR policy the threshold is set each round at the 99th percentile of a calibration sample of human texts. Table S12 makes that sample small (500 or 200 texts instead of 2,000), unrepresentative (G2 is 5% or 40% of it instead of 20%) or stale (drawn from the populations 3 or 10 rounds earlier). Small samples raise the realised false-positive rate, because a new text exceeds the empirical 99th percentile of k texts with probability of about $(k - \lfloor 0.99(k-1)\rfloor)/(k+1)$, about 1.5% for $k = 200$. Under-representing G2, whose scores are higher, lowers the threshold and raises the rate; over-representing it does the opposite. A stale sample lowers the rate, because writers have since moved away from the flagged region. In every configuration G2 still accumulates more false flags than G1, by a similar margin.

S2 Real-Text Protocol and Results

To describe round-0 accuracy and group error rates, we use 200 random 50/25/25 splits into training, calibration and evaluation sets, with the threshold set on the calibration set; because the splits reuse the same texts, the bracketed percentiles in Table S13 are split-variability intervals, not confidence intervals. To run the loop, we deploy a cross-fitted detector: the corpus is divided into five folds, a model fitted on four folds scores the fifth, and the threshold is set at 10% of these out-of-fold scores on human texts. Features are standardised inside each fold model using only its training texts, and a rewritten text keeps the fold of the text it replaces, so it is scored by a model that saw neither version. In round 1 the *frozen* detector is the round-0 detector unchanged, and the *retrained* detector a new cross-fitted detector fitted to the rewritten corpus with its own threshold. With 179 to 394 human texts, a 1% rate would rest on two to four flagged texts, which is why we use 10%.

The threshold is estimated from the same corpus that is flagged, so it is not an operating point validated on an independent population, and the five fold models' score scales need not coincide (out-of-fold human score standard deviations range from 2.4 to 3.0 and from 1.6 to 2.9). Replacing the pooled threshold by fold-specific ones changes the flag on 10 of 394 and 8 of 179 human texts, moves the non-native false-positive rate from 18.7% to 19.8% and from 16.5% to 19.8%, and leaves 17 of 18 and 18 of 18 flagged non-native essays flagged after polishing. Published rewrites exist only for the non-native essays and the GPT-3 texts, so the 23 and 3 native essays flagged in round 0 stay unchanged, and the check does not compare adaptation between groups. The detector's high AUROC should not be read as typical of task-matched texts, since the corpus also

Table S8: Deployment requirements and failure modes of the operating policies.

Policy	Requires	Fails when	Where
Recalibration to a fixed FPR	A labelled sample of human texts to set the threshold	By design it keeps the burden at α and makes writers' self-protection zero-sum; the loss of utility goes unnoticed unless it is audited separately	Props. 4.2, 4.3, 4.4
Fixed threshold	For the guarantee, scores equal to the equal-prior posterior, which ordinary probability calibration does not ensure; in practice, a stable training prior or recalibration on balanced labelled data	The training prior drifts, scores are rescaled, or the model is miscalibrated	Prop. 4.4(2); Supp. S1.8
FPR cap	A labelled sample of human texts to check the cap	Binds only while the detector separates the classes; if the operator lowers the threshold to the cap, it becomes recalibration	Secs. 4.3, 5.6
Fixed alert budget	An assumed share of machine text among submissions	As detection declines, its flags move from machine text to writers	Secs. 4.3, 5.6
Withdrawal after a precision audit	Correctly labelled audit texts that represent current submissions and current evasion; a prevalence that is not overestimated; for a confidence-bound audit at a 1% operating point, about 2,000 human texts	The prevalence is overestimated; audit machine texts are easier to detect than real submissions; labels are wrong; the audit is too small to bound the false-positive rate, in which case a confidence-bound audit withdraws a detector whose flags are still mostly correct	Prop. 4.4(3), Cor. 4.5; Secs. 4.4, 5.6; Supp. S1.9
Any policy	Group labels for reporting rates by group; records of what follows a flag	Without them, disparity and harm cannot be measured	Sec. 7.3

Table S9: Withdrawal with the audit drawn from the evaluation sample and with an independent audit sample (20 seeds). Rounds deployed are reported as mean (s.d.; range) across seeds; false flags are cumulative per writer.

Parameters	Rule	Rounds deployed		G2 false flags		G1 false flags	
		evaluation	independent	eval.	indep.	eval.	indep.
Default	$\pi = 0.1$	8.6	8.4 (1.0; 6–10)	0.24	0.24	0.05	0.05
	$\pi = 0.05$	4.2	4.6 (0.6; 3–5)	0.14	0.16	0.02	0.02
	$\pi = 0.25$	24.3	26.0 (4.5; 14–33)	0.44	0.47	0.21	0.23
	π: 0.05 → 0.3	6.3	6.2 (1.5; 4–9)	0.20	0.20	0.03	0.03
	AUROC < 0.75	16.4	16.3 (1.2; 14–19)	0.36	0.36	0.13	0.13
	audit of 200 per class	6.1	6.6 (2.0; 3–11)	0.19	0.21	0.03	0.04
Anchored	$\pi = 0.1$	1.6	1.5 (0.6; 1–3)	0.07	0.08	0.00	0.00
	audit of 200 per class	1.6	1.6 (0.8; 1–3)	0.07	0.07	0.00	0.00
	AUROC < 0.75	9.2	9.4 (0.9; 7–11)	0.45	0.46	0.01	0.01

Table S10: Effect of human adaptation on final AUROC under a TPR-scaled and a constant evasion step (20 seeds). Differences are response minus no response, paired by seed, with 95% t-intervals.

Parameters	Evasion step	Final AUROC			False flags, none → response	
		none	response	difference	G2	G1
Anchored	scaled with TPR	0.528	0.500	−0.028 [−0.031, −0.025]	2.75 → 0.99	0.10 → 0.54
	constant, $c = 0.018$	0.503	0.501	−0.002 [−0.004, −0.001]	2.32 → 1.09	0.21 → 0.51
	constant, $c = 0.031$	0.507	0.506	−0.001 [−0.003, +0.000]	1.91 → 0.95	0.31 → 0.55
Default	scaled with TPR	0.536	0.607	+0.070 [+0.066, +0.074]	1.89 → 0.82	0.32 → 0.58
	constant, $c = 0.042$	0.501	0.680	+0.179 [+0.170, +0.187]	1.76 → 0.83	0.35 → 0.59
	constant, $c = 0.078$	0.504	0.501	−0.003 [−0.005, −0.001]	1.49 → 0.80	0.42 → 0.59

differs in genre, source and prompt. A plausible explanation of the opposite movements is that polishing raises lexical sophistication, which the stylometric detector associates with GPT-3 text, while

Table S11: Alternative response functions (20 seeds). The G2 − G1 difference in cumulative false flags is paired by seed.

Configuration	Final AUROC	Mean FPR	False flags G1	False flags G2	G2 − G1 flags [95% interval]	Drift G1	Drift G2	Final FPR G1	Final FPR G2
Default (linear)	0.61	1.05%	0.58	0.82	+0.24 [+0.23, +0.24]	0.94	1.73	1.22%	1.16%
Threshold, 2×	0.53	1.05%	0.62	0.69	+0.08 [+0.05, +0.10]	0.34	0.50	1.30%	0.64%
Threshold, 5×	0.53	1.06%	0.43	1.45	+1.02 [+0.93, +1.11]	0.43	0.16	1.04%	1.63%
Saturating, cap 2×	0.61	1.06%	0.54	1.01	+0.47 [+0.45, +0.48]	0.71	1.49	1.21%	1.23%
Saturating, cap 1×	0.61	1.05%	0.46	1.33	+0.87 [+0.84, +0.91]	0.37	1.11	1.15%	1.49%
Partial awareness (25%)	0.56	1.05%	0.49	1.22	+0.73 [+0.72, +0.75]	0.16	0.58	1.18%	1.24%
Anchored, linear	0.50	1.05%	0.54	0.99	+0.44 [+0.43, +0.46]	0.16	1.50	1.26%	0.16%
Anchored, threshold 2×	0.50	1.06%	0.57	0.87	+0.29 [+0.27, +0.32]	0.29	1.11	1.28%	0.09%
Anchored, saturating, cap 2×	0.50	1.05%	0.40	1.55	+1.15 [+1.12, +1.18]	0.06	1.31	1.18%	0.43%

Table S12: Imperfect recalibration under the fixed-FPR policy (20 seeds). The G2 − G1 difference in cumulative false flags is paired by seed.

Configuration	Final AUROC	Mean FPR	False flags G1	False flags G2	G2 − G1 flags [95% interval]	Drift G1	Drift G2	Final FPR G1	Final FPR G2
Default (2,000 texts)	0.61	1.05%	0.58	0.82	+0.24 [+0.23, +0.24]	0.94	1.73	1.22%	1.16%
500 texts	0.60	1.20%	0.67	0.91	+0.24 [+0.24, +0.25]	1.17	1.96	1.23%	1.25%
200 texts	0.57	1.48%	0.84	1.09	+0.25 [+0.25, +0.26]	1.60	2.40	1.53%	1.33%
G2 share 5%	0.60	1.11%	0.62	0.86	+0.24 [+0.24, +0.25]	1.05	1.84	1.19%	1.18%
G2 share 40%	0.62	0.98%	0.54	0.77	+0.24 [+0.23, +0.24]	0.80	1.59	1.21%	1.23%
3 rounds stale	0.62	0.85%	0.46	0.69	+0.23 [+0.23, +0.24]	0.63	1.41	0.93%	0.84%
10 rounds stale	0.62	0.64%	0.34	0.57	+0.23 [+0.22, +0.23]	0.31	1.10	0.70%	0.64%
Anchored, 2,000 texts	0.50	1.05%	0.54	0.99	+0.44 [+0.43, +0.46]	0.16	1.50	1.26%	0.16%
Anchored, 200 texts	0.51	1.51%	0.84	1.15	+0.31 [+0.29, +0.32]	0.24	1.64	1.33%	0.34%
Anchored, 3 rounds stale	0.50	0.85%	0.41	0.91	+0.50 [+0.49, +0.51]	0.08	1.39	1.12%	0.31%

Table S13: One round of detection, rewriting and retraining with a stylometric detector at a 10% pooled FPR.

	All native	8th-grade only
R0 AUROC, held-out	0.98	0.99
R0 FPR, native, held-out	7.8% [1.3, 17.1]	4.9% [0.0, 18.2]
R0 FPR, non-native, held-out	19.1% [4.3, 35.0]	19.1% [4.3, 39.1]
R0 FPR, native, deployed	7.6%	3.4%
R0 FPR, non-native, deployed	18.7%	16.5%
Non-native flagged and polished	17	15
Native flagged (no rewrite)	23	3
Mean score change after polishing	+3.17	+4.86
Still flagged after polishing	16 of 17	15 of 15
R1 AUROC, held-out	0.97	0.98
R1 FPR, non-native, held-out	19.3% [4.3, 34.8]	17.3% [4.2, 39.1]
R1 FPR, non-native, frozen	17.6%	16.5%
R1 FPR, non-native, retrained	20.9%	15.4%

the commercial scores are consistent with a response to the higher perplexity of polished text; we have not tested the latter.

Leave-corpus-out evaluation. Table S14 reports the leave-corpus-out tests of Section 6.3 of the main paper. Each detector is trained with one sub-corpus held out entirely, its threshold is set at a 10% false-positive rate on its own training human texts, and it is evaluated only on the held-out sub-corpus and on the TOEFL essays. Held-out machine text exists only for the college-essay and abstract sub-corpora, so the Hewlett design reports false-positive rates only.

S3 Reproducibility Map

Table S15 links each figure and table to the script, the configuration and the stored result file that produce it. All scripts are in the supplementary archive; `run_all.sh` runs them in order and compares every regenerated result file with `checksums.sha256`. `make_tables.py` and `make_figures.py` print the reported values from the stored files.

Table S14: Leave-corpus-out evaluation of the stylometric detector at a 10% false-positive rate on its training human texts.

Held out	Training corpus (human vs. machine)	CV AUROC in domain	Held-out AUROC	Held-out TPR	FPR, held-out native	FPR, TOEFL before → after	TOEFL essays with higher score
College essays	Hewlett, abstracts vs. GPT-3 abstracts	0.998	0.69	3%	0%	4% → 24%	96%
Abstracts	Hewlett, college vs. GPT-3 college	0.984	0.96	10%	0%	63% → 82%	85%
Hewlett essays	college, abstracts vs. both GPT-3 sets	0.993	–	–	64%	79% → 93%	85%

Table S15: Reproducibility map. Configurations are dictionaries in `simulation/arms_race_sim.py` unless stated otherwise.

Item	Script	Configuration	Result file
Figure 2, Table 3	`make_figures.py` (analytical)	–	–
Figures 3–5, Tables 4–5	`arms_race_sim.py`	`MAIN`	`results.json`
Tables 6–7	`arms_race_sim.py`	`CALIBRATED`	`calibrated.json`
Table 8	`arms_race_sim.py`, `nonlinear_sim.py`	`RESPONSES`, `SENSITIVITY`, `MAIN`	`extensions.json`, `sensitivity.json`, `results.json`, `nonlinear.json`
Table 9	summary of Tables 7, 8 and S3, S4, S10, S11	–	–
Table 10 (fixed FPR, fixed threshold, point-estimate withdrawal)	`arms_race_sim.py`	`POLICIES`, `CALIBRATED`	`extensions.json`, `calibrated.json`
Table 10 (FPR cap, fixed alert budget, confidence-bound withdrawal)	`policy_revision_runs.py`	`CONFIGS` in script	`policy_revision.json`
Table 11	`experiment/published_detectors.py`	–	`published_detectors.json`
Section 6.2, Table S13	`experiment/run_experiment.py`, `crossfit_check.py`	`--native` all or `hewlett`	`stylometric_all_native.json`, `stylometric_hewlett_native.json`, `crossfit_check.json`
Table 12	`experiment/matched_check.py`	–	`matched_check.json`
Section 6.3, Table S14	`experiment/leave_corpus_out.py`	–	`leave_corpus_out.json`
Table S1, Figure S1	`arms_race_sim.py`	`SENSITIVITY`, `SWEEP`	`sensitivity.json`, `sweep.json`
Section S1.3	`arms_race_sim.py`	`ROBUSTNESS`	`robustness.json`
Tables S2–S3	`arms_race_sim.py`	`ABLATIONS`, `RESPONSES`	`extensions.json`
Table S4	`nonlinear_sim.py`	in script	`nonlinear.json`
Table S5	`anchors.py`, `arms_race_sim.py`	`anchor_configs()`	`anchors.json`, `extensions.json`
Table S6, Section S1.8	`arms_race_sim.py`, `fixed_threshold_check.py`	`POLICY_ROBUSTNESS`; in script	`policy_robustness.json`, `fixed_threshold_check.json`
Table S7	`prevalence.py`	in script	`prevalence.json`
Tables S9–S10	`audit_evasion_checks.py`	in script	`audit_evasion_checks.json`
Tables S11–S12	`response_calibration_checks.py`	in script	`response_calibration_checks.json`

References

[1] George Forman. 2008. Quantifying counts and costs via classification. *Data Mining and Knowledge Discovery* 17, 2 (2008), 164–206.

[2] Marco Saerens, Patrice Latinne, and Christine Decaestecker. 2002. Adjusting the Outputs of a Classifier to New a Priori Probabilities: A Simple Procedure. *Neural Computation* 14, 1 (2002), 21–41.